

Derivatives

1	$(c)' = 0$
2	$(x^n)' = nx^{n-1}$
3	$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
4	$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$
5	$(e^x)' = e^x$
6	$(a^x)' = a^x \ln a$
7	$(\ln x)' = \frac{1}{x}$
8	$(\log_a x)' = \frac{1}{x \ln a}$
9	$(\sin x)' = \cos x$
10	$(\cos x)' = -\sin x$
11	$(\tan x)' = \frac{1}{\cos^2 x}$
12	$(\cot x)' = -\frac{1}{\sin^2 x}$
13	$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$
14	$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$
15	$(\arctan x)' = \frac{1}{1+x^2}$
16	$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$
17	$(\sinh x)' = \cosh x$
18	$(\cosh x)' = \sinh x$
19	$(\tanh x)' = \frac{1}{\cosh^2 x}$
20	$(\coth x)' = -\frac{1}{\sinh^2 x}$
21	$(\operatorname{arsinh} x)' = \frac{1}{\sqrt{x^2+1}}$
22	$(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2-1}}$
23	$(\operatorname{artanh} x)' = \frac{1}{1-x^2}$
24	$(\operatorname{arcoth} x)' = \frac{1}{1-x^2}$

Integrals

$$\boxed{1} \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\boxed{2} \quad \int \frac{dx}{x} = \ln |x| + C$$

$$\boxed{3} \quad \int e^x dx = e^x + C$$

$$\boxed{4} \quad \int a^x dx = \frac{a^x}{\ln a} + C$$

$$\boxed{5} \quad \int \sin x dx = -\cos x + C$$

$$\boxed{6} \quad \int \cos x dx = \sin x + C$$

$$\boxed{7} \quad \int \frac{dx}{\cos^2 x} = \tan x + C$$

$$\boxed{8} \quad \int \frac{dx}{\sin^2 x} = -\cot x + C$$

$$\boxed{9} \quad \int \tan x dx = -\ln |\cos x| + C$$

$$\boxed{10} \quad \int \cot x dx = \ln |\sin x| + C$$

$$\boxed{11} \quad \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$\boxed{12} \quad \int \frac{dx}{1+x^2} = \arctan x + C$$

$$\boxed{13} \quad \int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C$$

$$\boxed{14} \quad \int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\boxed{15} \quad \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C$$

$$\boxed{16} \quad \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\boxed{17} \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C$$

$$\boxed{18} \quad \int \sinh x \, dx = \cosh x + C$$

$$\boxed{19} \quad \int \cosh x \, dx = \sinh x + C$$

$$\boxed{20} \quad \int \frac{dx}{\cosh^2 x} = \tanh x + C$$

$$\boxed{21} \quad \int \frac{dx}{\sinh^2 x} = -\coth x + C$$

$$\boxed{22} \quad \int \tanh x \, dx = \ln \cosh x + C$$

$$\boxed{23} \quad \int \coth x \, dx = \ln |\sinh x| + C$$

$$\boxed{24} \quad \int \frac{dx}{x \ln x} = \ln |\ln x| + C$$